Machine Learning tools to assist the analysis of complex rotational spectra of chemical mixtures

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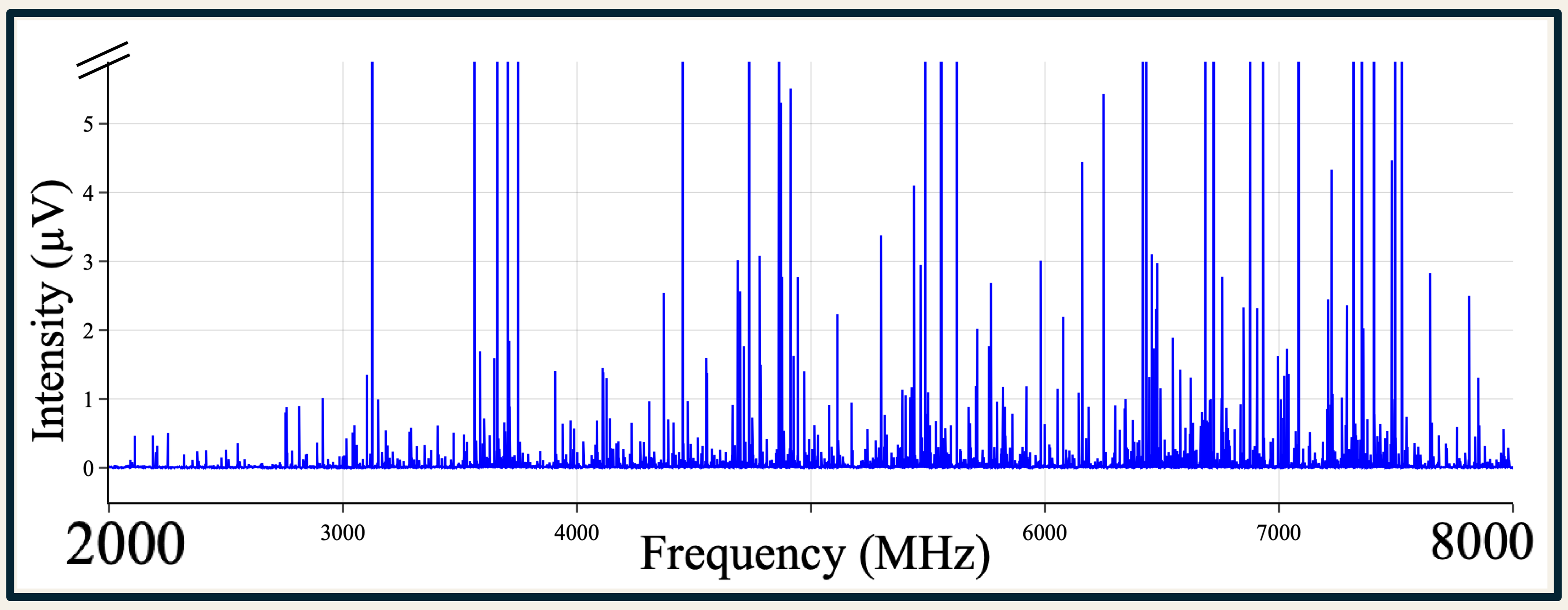
Introduction

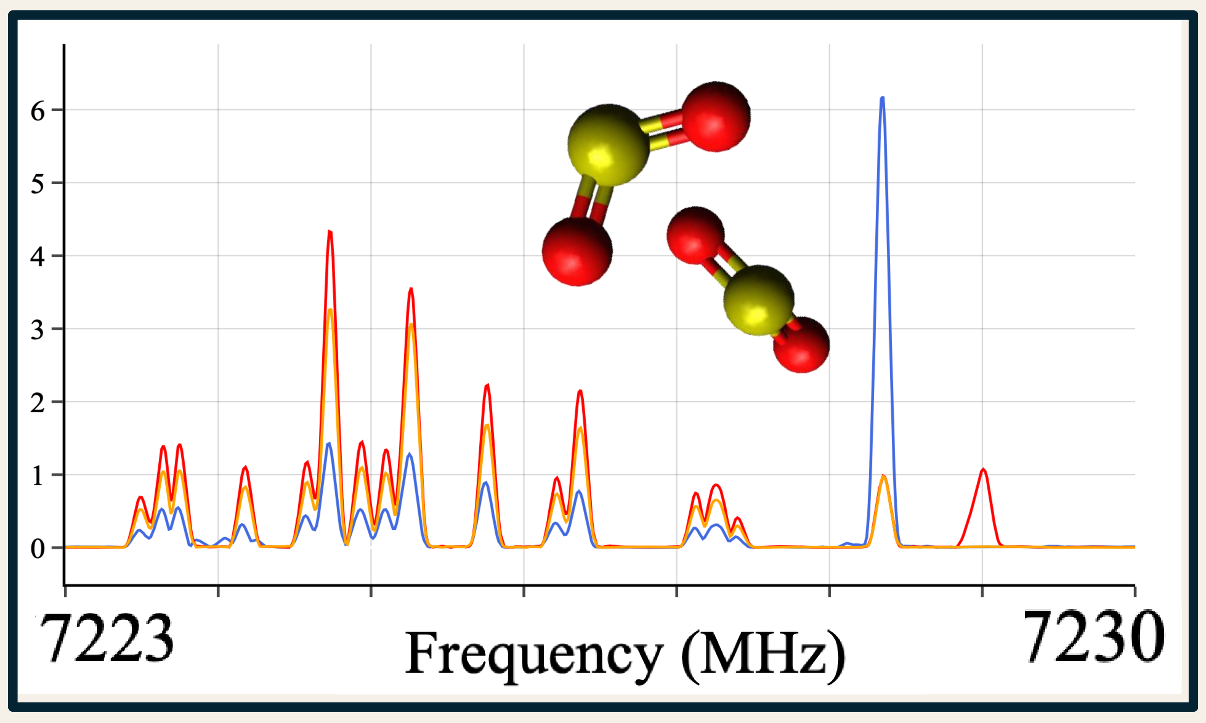
The possibility of averaging millions of spectra thanks to Broadband chirped pulse Fourier Transform microwave spectroscopy, makes rotational analysis challenging involving thousands of transitions lines. New methods have arised to faciltate the analysis of the molecular spectra, such as frequency-based methods [1]. Within this study we focus on intensity-based methods for chemical mixtures ($SO_2/H_2O/D_2O$), where we use the unique response of each chemical species to changes in composition. We explore new approaches using Machine Learning tools to disentagle rotational spectra.

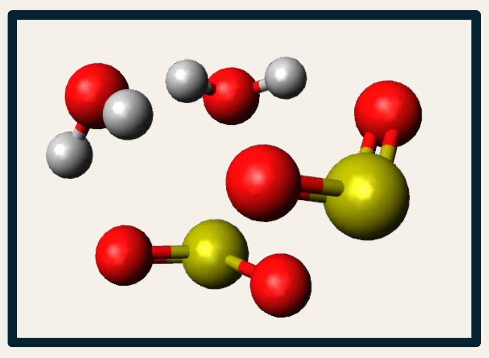
Experimental

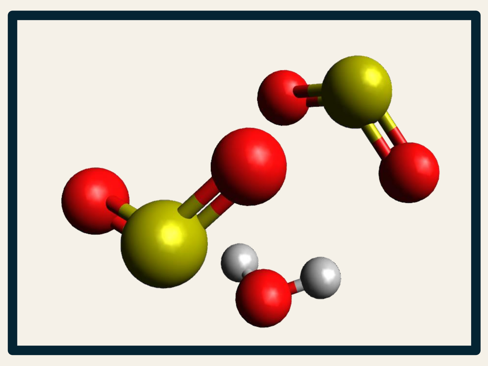
We measured rotational spectra of three different mixtures using Broadband CP-FTMW spectrometer placed in Universidad de Valladolid.

SO_2 pure | $+H_2O$ | $+D_2O$



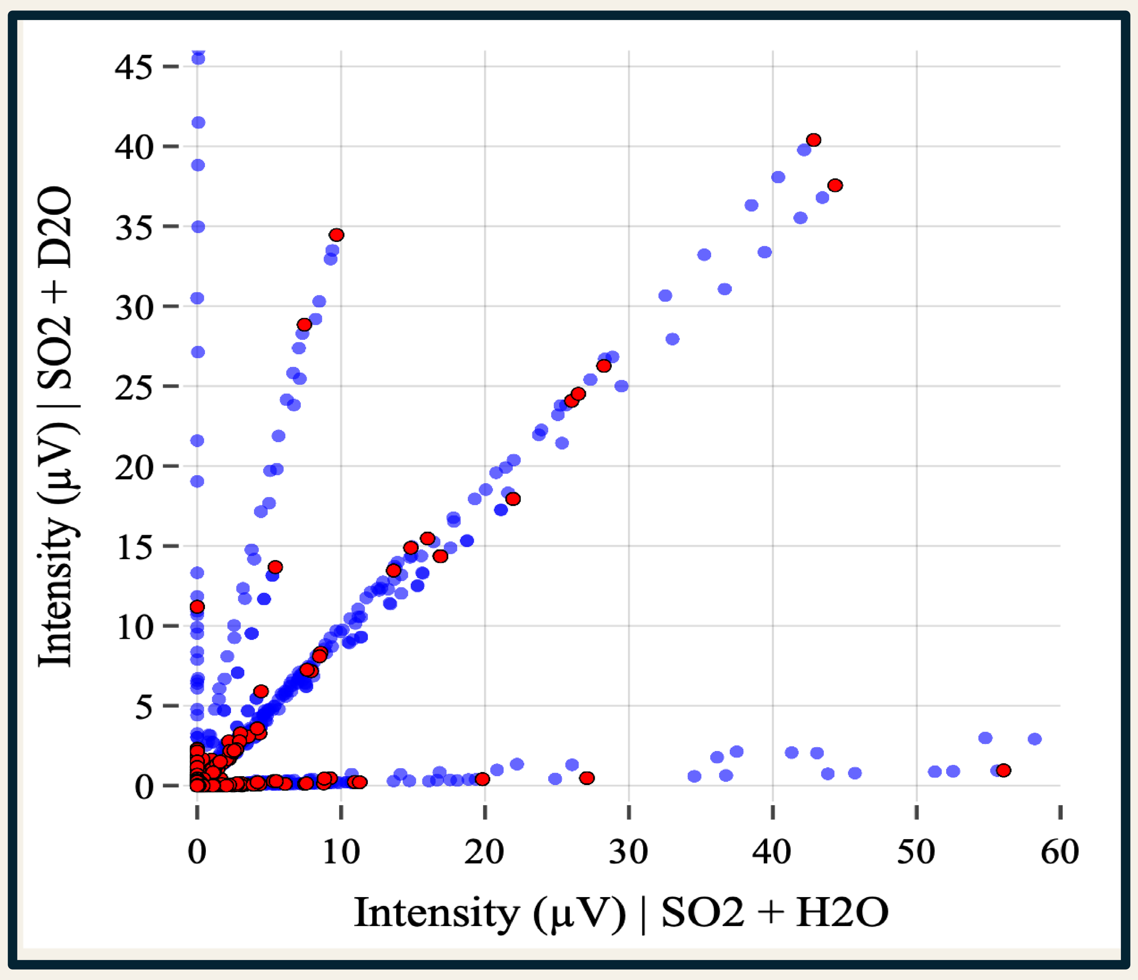


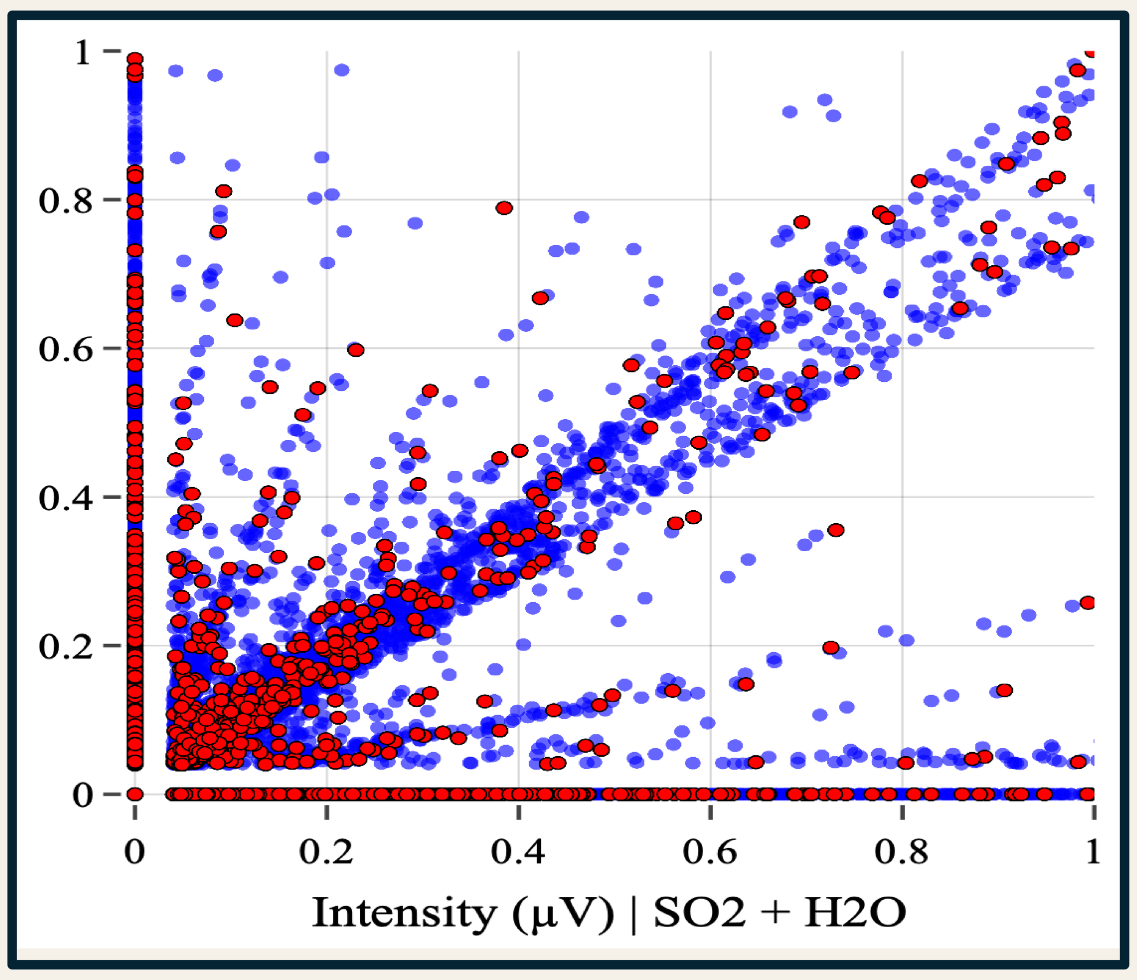




Methods

By facing intensities of the spectra of two mixtures with different composition [2], we detect linear patterns called "rays". Each ray corresponds to a unique response to changes in composition and therefore to a specific chemical specie. We developed different unsupervised clustering Machine Learning algorithms to isolate each ray and assign each cluster to a chemical species of SO_2 / H_2O. We focus on the best perfomance achieved by RANSAC algorithm.





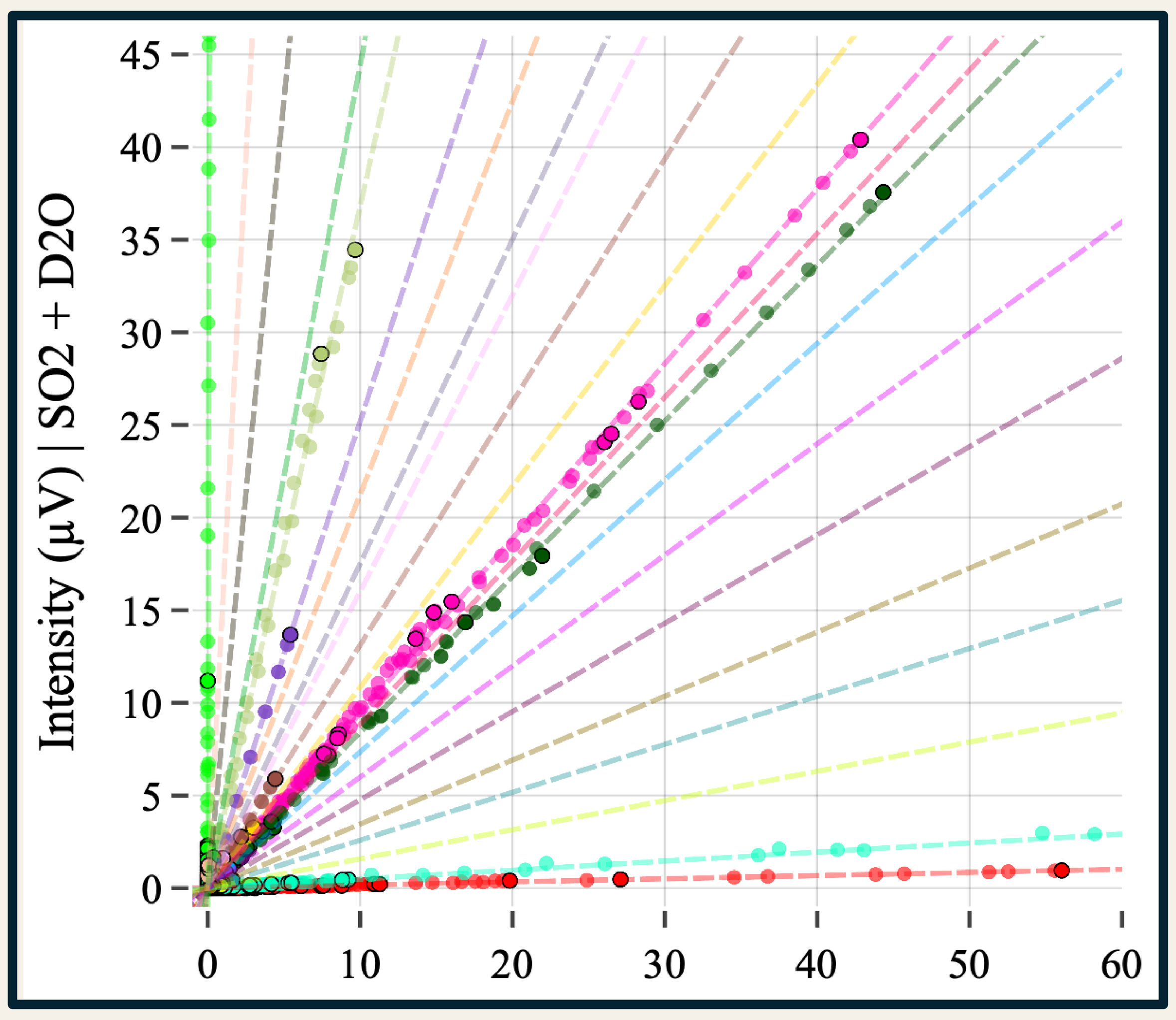
Modified RANSAC Algorithm

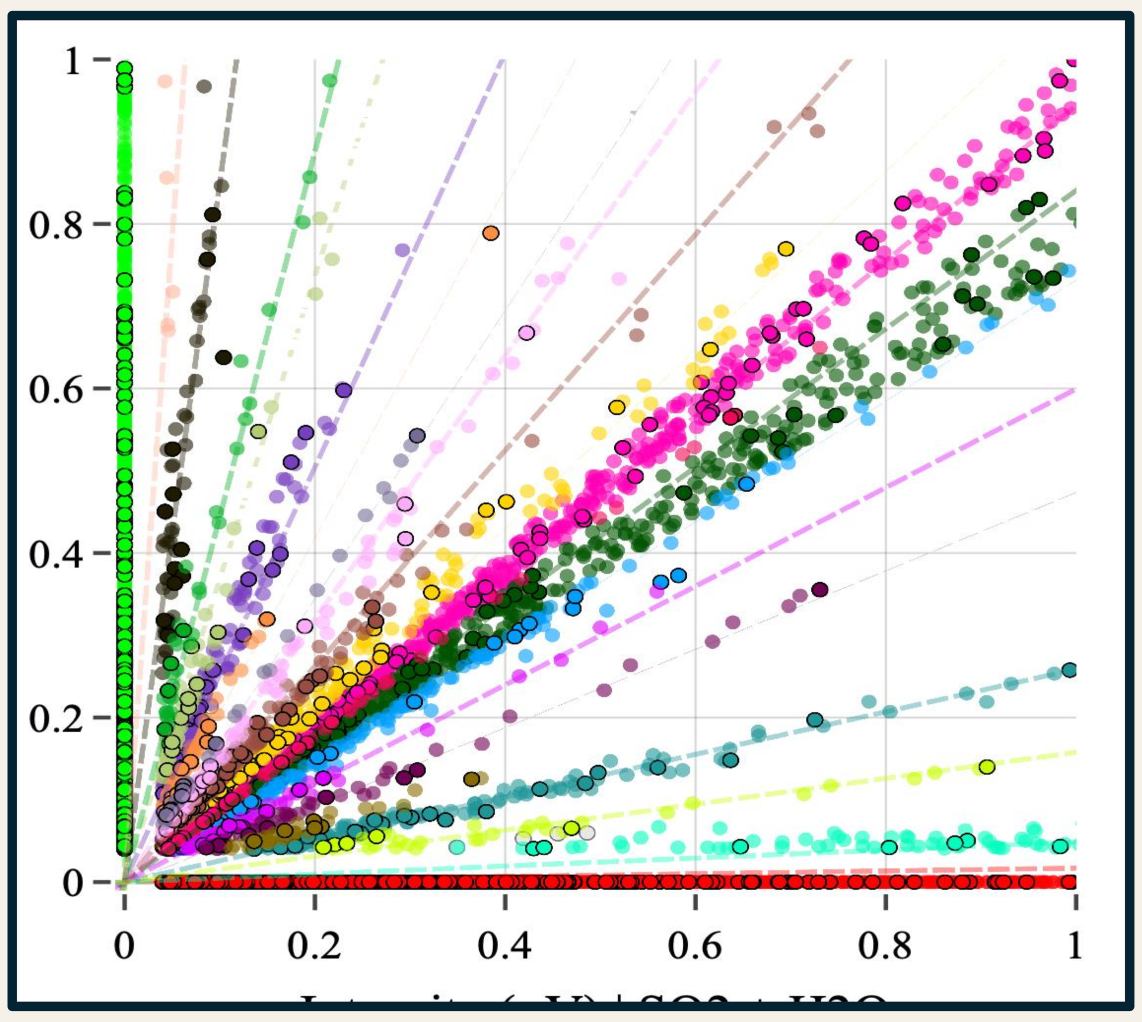
1. Iterates over all possible linear regressions forcing the intercept to origin.
2. Scores regressions according to the number of inliers within an angular threshold.
3. Clusters all inliers within the best regression.
4. Removes this cluster and repeats the process.

Hyperparameters

- Minimum datapoints used per regression = 2
- Maximum number of clusters = 20
- Maximum iterations = 10 000
- Angular threshold = 0.02 rads
- Threshhold growth = +30% / cluster
- Maximum Threshold = 0.04 rads

Results of RANSAC model



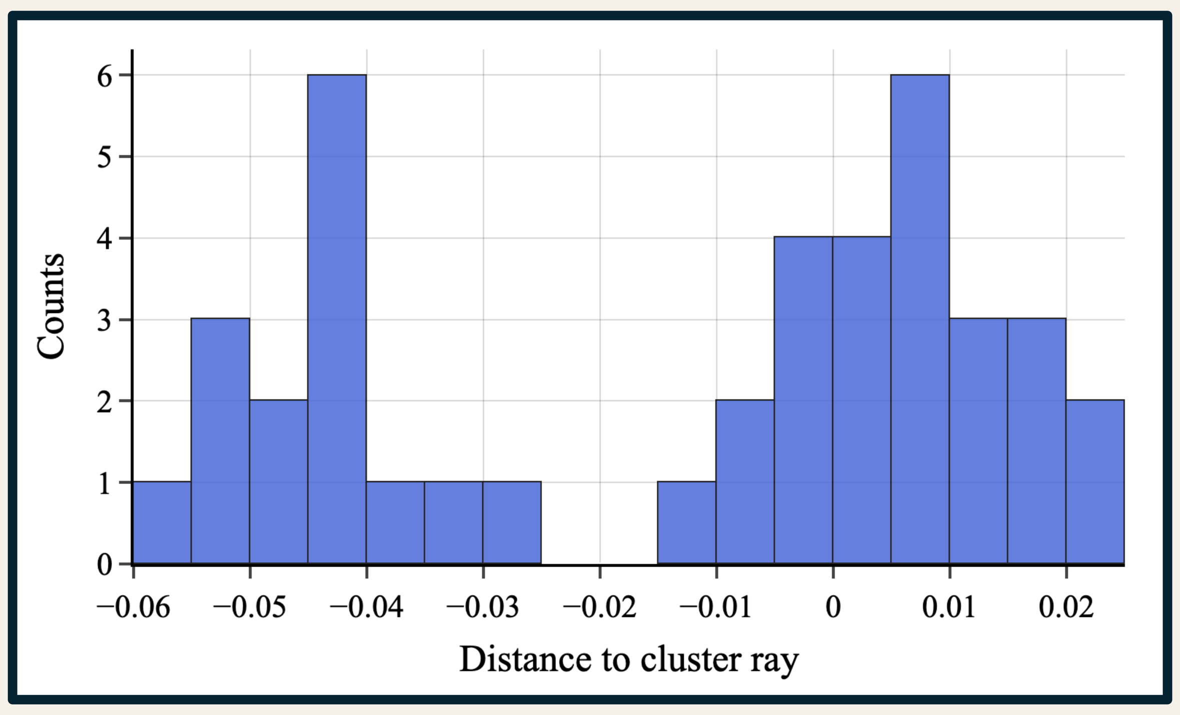


Discussion

After trying other model's algorithm like DBSCAN, FLEMIX and GMM we consider RANSAC algorithm to be best suited due to its hyperparameter flexibility. RANSAC shows great results. However it is sensitive to high densities of data and distance threshold. Hyperparameters adjustment is always requiered for specific problems.

How to deal with these drawbacks:

- Work only with maximums as datapoints to avoid overfitting from intense signals.
- Analyze distributions of each datapoint distance to each ray slope.



References

[1] N. A. Seifert et al., Mol. Spectrosc. 321, 13-21 (2015).
[2] K. C. Gilbert et al., J. Phys. Chem. A. 129 (45), 10393-10403 (2025).

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